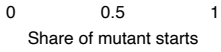
a **All historical serial designs: transfer 8**

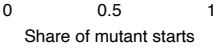
Design (L bytes)	Controls alive	Scored sites
Z80 pushers		
Pusher L16	3/256	0
Pusher L50	0/256	0
Pusher L64	0/256	0
Evolved Z80 closers		
Return L16	256/256	0
Block-copy L20	256/256	0
Block-copy L32	256/256	0
Open jump L50	0/256	0
Evolved Z80 transmitters		
Lethal T L16	256/256	9
T L16 seed 6006	256/256	10
T L20 seed 6003	256/256	9
T L20 seed 6004	256/256	12
Constructed 8080 controls		
Open 8080 pusher L32	9/256	0
Copier L32, variant 1	256/256	19
Copier L32, variant 2	256/256	20



Dark grey lost; light grey alive noncarrier; teal alive carrier
Carrier = tape contains centred cyclic mutant 5-byte window
Site: ≥half identifiable tested values have excess carrier share ≥0.5
Excess = carriers/survivors (mutant) − carriers/survivors (WT)

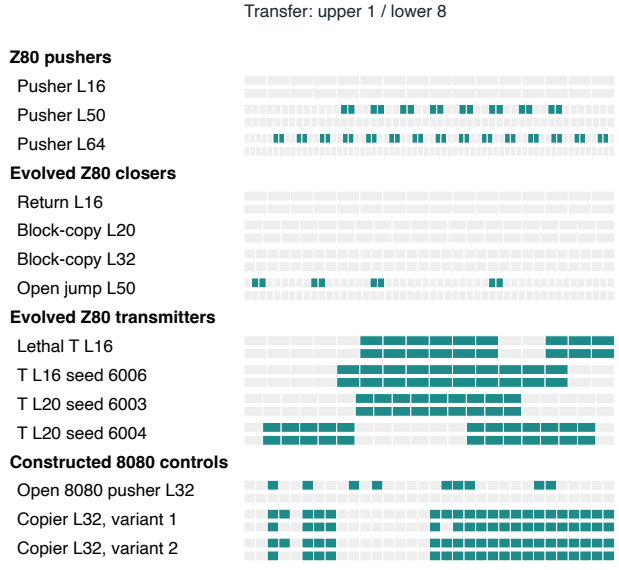
b **Branching transfer-8 outcomes: U / K**

Design (L bytes)	Controls alive	Scored sites
Pusher L16 U	48/64	0
Pusher L16 K	48/64	0
Pusher L64 U	8/64	0
Pusher L64 K	28/64	0
Return L16 U	64/64	0
Block-copy L32 U	64/64	0
T L20 seed 6003 U	64/64	9



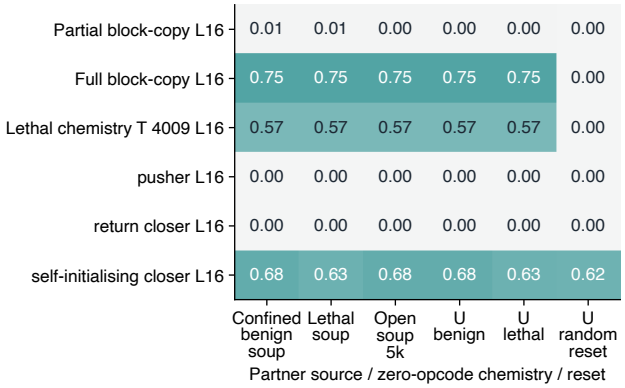
Grey: no surviving lineage; pale: alive, <50% members carry
Teal: alive, ≥50% members carry
Carriage = members containing mutant 5-byte window / all members
Site: ≥half identifiable tested values have excess carrier share ≥0.5
Excess as a; U = uniform-byte partners; K = evolved-soup partners
32 values×4 starts/site; WT = unmutated control (64)

c **Complete serial site maps: 1 / 8**



Upper: transfer 1; lower: transfer 8
Teal = scored site; pale = tested, unscored site
Site: ≥half identifiable tested values have excess carrier share ≥0.5
Same excess definition as a; tracks span bytes 0 to L−1

d **All partner-environment carriage tests**



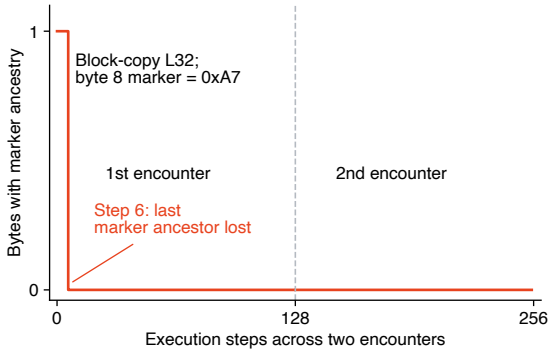
Transfer-8 carriers / all mutant starts; U = uniform-byte partners
All 6 parents L16: 16×255×16 = 65,280 starts/design; WT 256
Benign zero executes; lethal zero halts; open soup at 5,000 updates

e **Identifiable alternatives and controls**

Design	Identifiable / tested alternatives	WT starts
Z80 pushers		
Pusher L16	4080/4080	256
Pusher L50	1600/1600	256
Pusher L64	2048/2048	256
Evolved Z80 closers		
Return L16	4080/4080	256
Block-copy L20	5100/5100	256
Block-copy L32	8160/8160	256
Open jump L50	1600/1600	256
Evolved Z80 transmitters		
Lethal T L16	4080/4080	256
T L16 seed 6006	4080/4080	256
T L20 seed 6003	5100/5100	256
T L20 seed 6004	5100/5100	256
Constructed 8080 controls		
Open 8080 pusher L32	8160/8160	256
Copier L32, variant 1	8160/8160	256
Copier L32, variant 2	8160/8160	256

Identifiable: marker window absent from WT; T = evolved transmitter
Alternative types = L byte positions × tested values
16 starts/value; pusher50/64 & jump50: 32 values, others 255

f **Ordered ancestry: two full encounters**



Counted unit: bytes with byte-8 ancestry; all-zero byte partners
Step 6 reads address 4 (ancestry byte 0), writes 0x04 over 8
At execution step 128: offspring becomes parent; processor reset